



Letter to the Editor

Artificial intelligence-assisted CCTA for coronary stenosis detection: Diagnostic promise, methodological nuances, and directions for clinical translation**1. Introduction**

We read with great interest the recent *meta*-analysis by Yan et al., which comprehensively evaluates the diagnostic performance of coronary computed tomography angiography (CCTA)-based artificial intelligence (AI) systems for detecting $\geq 50\%$ coronary stenosis at both patient and vessel levels [1]. By rigorously adhering to PRISMA-DTA standards and disentangling patient-level from vessel-level evidence, the authors address a critical methodological limitation present in prior *meta*-analyses and provide valuable quantitative insight into the evolving role of AI-assisted CCTA in coronary artery disease (CAD) assessment.

The reported diagnostic performance is particularly noteworthy. At the patient level, pooled sensitivity reached 0.95 with an area under the curve (AUC) of 0.96, highlighting strong rule-out capability for clinically significant stenosis. Meanwhile, vessel-level analysis demonstrated a more balanced sensitivity (0.87) and specificity (0.89), with an AUC of 0.93, underscoring the potential of AI tools for lesion localization and procedural planning. These findings are consistent with earlier single-study and pooled analyses suggesting that deep learning-based CCTA interpretation can approach, or in selected contexts exceed, expert visual assessment [2,3].

At the same time, the persistently lower patient-level specificity (0.73) warrants further reflection. As the authors appropriately note, this trade-off likely reflects intentional threshold optimization favoring sensitivity in screening-oriented workflows. However, from a clinical decision-making perspective, false-positive patient-level classifications may lead to unnecessary downstream testing or invasive angiography, particularly in populations with intermediate pretest probability. Future work might therefore benefit from explicitly reporting operating-point-specific performance or decision-curve analyses to clarify how different threshold settings alter net clinical benefit across risk strata [4].

Another important contribution of this study lies in its explicit recognition that pooled performance reflects an aggregate of heterogeneous AI systems rather than the capability of any single model. Included studies varied substantially in architecture (e.g., InceptionV3 vs. U-Net-based networks), training datasets, and degrees of technical transparency, with several proprietary systems lacking detailed methodological disclosure. While *meta*-regression suggested that algorithm type did not significantly explain heterogeneity, the limited number of studies and underrepresentation of certain architectures constrain firm conclusions. Standardized reporting frameworks for AI in imaging, such as CLAIM or TRIPOD-AI, may help future investigations improve inter-study comparability and facilitate more granular synthesis [5].

The distinction between patient-level and vessel-level performance

also raises clinically meaningful questions regarding workflow integration. The superior sensitivity observed at the patient level supports AI deployment as a front-line triage or screening adjunct, whereas the higher specificity at the vessel level aligns more closely with interventional planning and targeted evaluation. Explicitly framing AI outputs within these complementary clinical roles—rather than as a monolithic diagnostic replacement—may enhance acceptance among clinicians and mitigate concerns regarding overdiagnosis or automation bias [6].

Several limitations acknowledged by the authors deserve continued emphasis. Despite including 1,506 patients and 2,896 vessels, the *meta*-analysis draws from only 11 primary studies, most of which were retrospective. Variability in reference standards (ICA, QCA, or expert consensus) introduces potential verification bias, and incomplete reporting of blinding and inclusion criteria in some studies may affect internal validity. Large-scale, prospective, multicenter evaluations with harmonized reference standards remain essential to confirm real-world performance and generalizability.

Finally, the broader implications of AI-assisted CCTA extend beyond stenosis detection alone. Emerging evidence suggests that AI-enhanced plaque characterization, functional assessment (e.g., CT-derived fractional flow reserve), and prognostic modeling for major adverse cardiovascular events may collectively offer more comprehensive risk stratification than luminal narrowing thresholds alone [7–9]. Integrating these multiparametric outputs into future diagnostic frameworks could help address the specificity bottleneck highlighted in the present study while aligning AI development with contemporary precision cardiology paradigms.

In summary, Yan et al. provide a timely and methodologically robust synthesis that strengthens the evidence base for AI-assisted CCTA in detecting $\geq 50\%$ coronary stenosis. By clearly delineating patient- and vessel-level performance and candidly discussing technical and clinical limitations, this work offers a valuable foundation for subsequent model-level validation and prospective clinical translation. Continued efforts toward standardization, transparency, and outcome-oriented evaluation will be crucial for realizing the full clinical potential of AI in coronary imaging.

Compliance with ethical standards*Research Involving human participants and/or animals*

This research did not require the involvement of human or animal subjects. Therefore, ethical approval for this study was not required according to local regulations and institutional policies.

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Data availability

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

CRediT authorship contribution statement

Weihao Cheng: Conceptualization, Data curation, Formal analysis, Writing – original draft, Writing – review & editing. **Zekai Yu:** Conceptualization, Writing – review & editing.

Informed consent

Not applicable.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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